



AI • IoT • Real-time Detection • Smart Alerts

SentinelEye

ML-Based Wild Animal Intrusion Detection System Using Raspberry Pi



Real-time
Detection



Instant
Alerts



SMS
Notification



Data
Logging



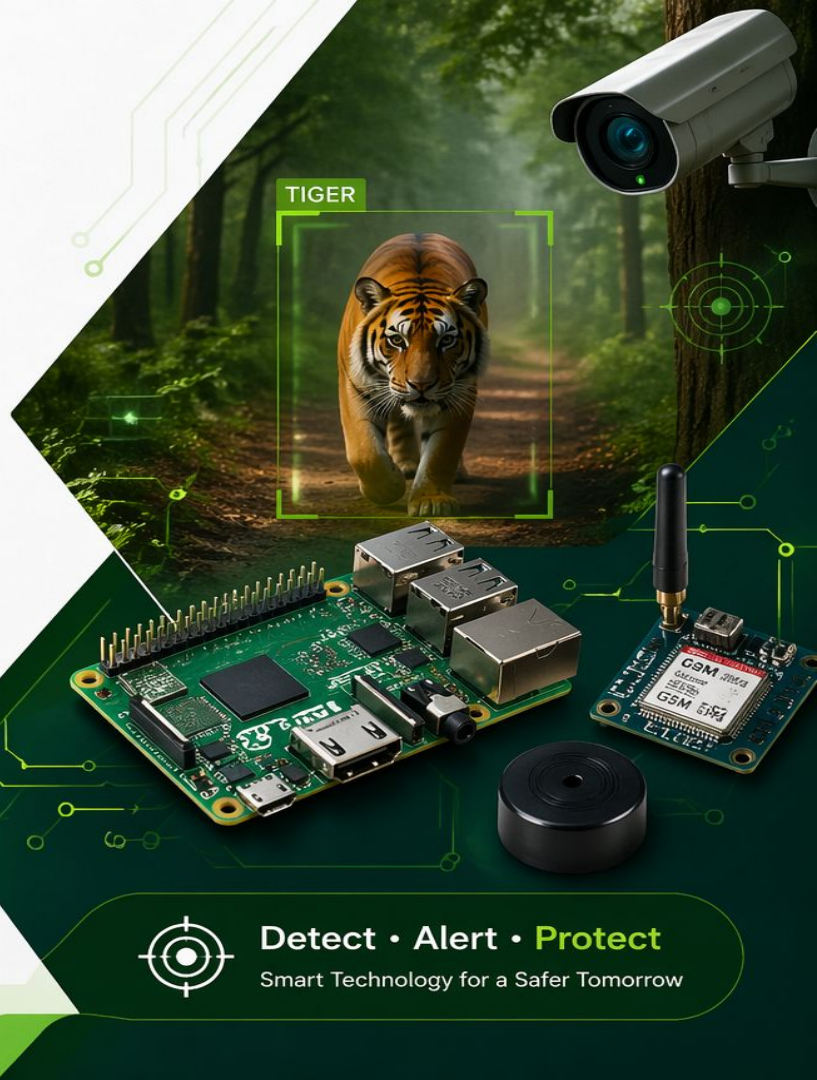
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Guided By

Prof. C.H. Patil



Detect • Alert • Protect

Smart Technology for a Safer Tomorrow

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1. Introduction

1.1 Overview:

- Wild animal intrusions are a serious threat not only to agriculture but also to public safety in urban and semi-urban areas.
- Such incidents occur in farmlands, residential colonies, city outskirts, schools, forest offices, and other public areas.
- These intrusions cause crop damage, financial loss, accidents, and even threats to human life.
- With the rise of AI, ML and IoT an automated system can be developed to detect animals in real-time, generate alerts, and prevent harm.

1.2 Need:

- Wild animal intrusions are increasing and causing safety risks.
- Farmers suffer major crop and financial losses.
- Human–animal conflicts threaten public safety.
- Existing methods are costly and inefficient.
- Manual monitoring is not practical in large areas.
- An automated, low-cost detection system is required.
- ML-based alerts enable quick preventive action.
- Raspberry Pi with AI/IoT offers a scalable solution.

1.3 Problem Statement:

Wild animal intrusions are increasingly reported in agricultural fields, city areas, schools, and forest regions, leading to crop damage, financial losses, accidents, and threats to human safety. Traditional solutions like fencing, guards, or CCTV are costly and ineffective in many cases. There is a need for an intelligent, real-time, and affordable system to detect animals, alert people instantly, and prevent conflicts.

1.4 Objectives:

- Detect and identify wild animals in agricultural and public areas.
- Achieve high model accuracy with minimal false alerts.
- Generate customized sound alerts based on the detected animal.
- Send instant SMS to nearby people and authorities with time + Farm/Area location.
- Store detection logs, images, and video clips locally or in the cloud for future analysis.
- Ensure applicability in farmlands, city outskirts, schools, and other vulnerable areas.

2. Literature Survey

Sr no.	Title	Authors	Publication & year	Pros	Cons
1	Enhancing Farm Security System With AI-Power-Driven Animal Intrusion Detection Mechanism.	Akshay Saxena, Aniket Shisodia, Divya Upadhyay	IEEE, 2025	Real-time alerts, automated monitoring, scalable	Farm-only focus, lighting/weather issues, simpler models
2	Animal Intrusion Detection in Farming Area using YOLOv5 Approach	Normaisharah Mamat, Mohd Fauzi Othman, Fitri Yakub	IEEE, 2022	High accuracy (~94%), detects multiple animals, state-of-the-art DL model	Limited to 4 animal types, high computation, less field testing
3	Implementation of a Wild Animal Intrusion Detection Model Based on Internet of Things	Prabhat Kumar Panda, Cherlopalli Srujan Kumar, Bommu Sai Vivek, Shashi Kant Dargar, Masura Balachandra	IEEE, 2022	Low-cost IoT solution, image capture + alerts, adaptable	Sensor placement issues, less accuracy, limited coverage

3. Proposed System Architecture

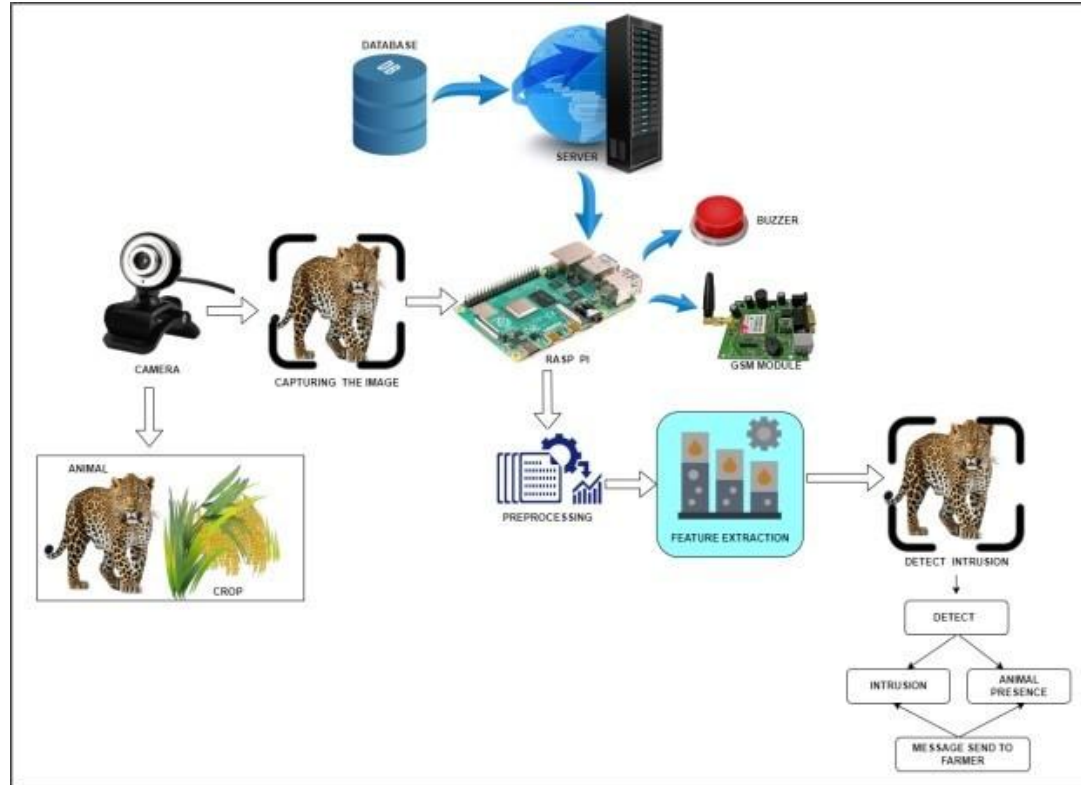


Figure 1: System Architecture of SentinelEye

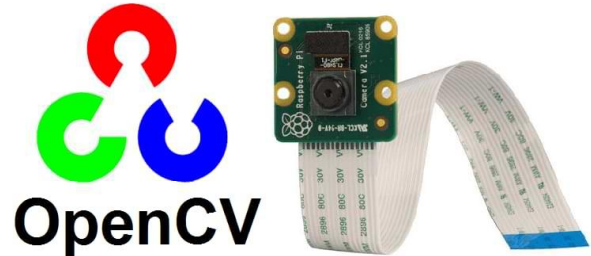
- **Central Unit:** Raspberry Pi (Raspbian OS) runs a deep learning model (ResNet-18) using PyTorch along with OpenCV for real-time image processing and detection.
- **Detection Process:** A pre-trained ResNet-18 CNN model (trained on ImageNet dataset) continuously processes frames from a live camera feed and identifies animals (elephants, leopards, boars, Tigers, Lions).
- **Efficiency:** The system processes live video frames in real-time using optimized inference (no training on device). Efficient enough to run on Raspberry Pi by using: Pretrained lightweight model (ResNet-18) Frame-by-frame inference (no heavy video storage).
- **Accuracy & Reliability:** Differentiates **harmful intruders** from harmless animals (dogs, cows) to avoid false alerts.

- **Response System:** On detection, triggers sound alarms (siren/buzzer) and sends SMS alerts via GSM with animal type, Field Location, and time.
- **Data Logging:** Field Location ensures precise tagging; detections and images are stored locally or on the cloud.
- **Design Goals:** Cost-effective, scalable, and reliable architecture with real-time alerts.
- **Applications:** Ideal for farms, highways, schools, and forest-edge areas to reduce crop damage and human-animal conflicts.

4. Details of Modules

Module 1: Image Acquisition Module

- Captures real-time video using camera (Pi Camera)
- Uses OpenCV (cv2.VideoCapture)
- Extracts frames continuously for processing
- **Purpose:** Provide live input to the detection system



Module 2: Image Preprocessing Module



- Converts image to RGB format
- Resizes image to **224×224 pixels**
- Normalizes pixel values using standard mean & std
- Uses Torchvision transforms
- **Purpose:** Prepare image for deep learning model input

Module 3: Animal Detection Module (ML Model)

- Uses **ResNet-18 CNN model** from PyTorch
- Pretrained on **ImageNet dataset**
- Performs **real-time image classification**
- Predicts object label (e.g., Leopard, Tiger, etc.)
- **Purpose:** Detect presence of animals in frame

Module 4: Alert & Notification Module

-  Activates buzzer using RPi.GPIO
-  Voice alert using espeak
-  Sends SMS via GSM module using pySerial
- **Purpose:** Provide real-time warning to users

5. Algorithm

- **Algorithm : ResNet-18 (Image Classification Algorithm)**

The ResNet-18 algorithm is a deep learning–based Convolutional Neural Network (CNN) used for image classification. It uses residual connections to overcome vanishing gradient problems and enables efficient feature extraction. In this project, the pretrained ResNet-18 model (trained on ImageNet dataset) is used to classify objects in real-time camera frames and identify animals.

- **Input**

1. Live video frames captured from Raspberry Pi camera.
2. Each frame is converted into an image format for processing

Preprocessing

- Convert image from BGR to RGB format
- Resize image to **224 × 224 pixels**
- Normalize using mean and standard deviation
- Convert image into tensor format for model input

Processing (Model Inference):

- Load pretrained **ResNet-18 model using PyTorch**
- Pass input image tensor to the model
- Model predicts class label from ImageNet dataset
- Extract predicted class (e.g., dog, cat, etc.)

Decision Making

- Check if detected label belongs to animal category
- Compare with predefined list of animals
- Avoid repeated alerts using time condition

Output / Action

If animal is detected:

- Activate buzzer using GPIO
- Generate voice alert using speaker (e.g., “*Leopard is detected*”)
- Send SMS alert using GSM module
- Store detection details (animal, time, location) in log file

6. S/W & H/W Requirements

6.1 Hardware Requirements:

Sr. no.	Hardware	Specifications	Purpose
1	Raspberry Pi 5	RAM 4/8 GB	Main processing unit
2	Raspberry Pi Power Supply	27W USB-C official one	Stable power supply for Pi 5
3	Camera	Pi (12MP Sony IMX708, autofocus)/ USB Camera	Video Streaming and object detection
4	Siren/ Speaker / Buzzer	Compatible	For alerts
5	GSM Module	SIM800L	SIM-based SMS alerts
6	GPS Module	NEO-6M GPS Module with EPROM	Location tagging
7	MicroSD card / PD	32/64 GB Class 10, UHS-I	OS + Storage

6.2 Software Requirements:

Sr No.	Category	Software / Library	Description
1	Operating System	Raspberry Pi OS (Raspbian)	Linux-based OS used to run all software on Raspberry Pi
2	Programming Language	Python 3.x	Main language used for coding the system
3	ML Framework	PyTorch	Loads pretrained ResNet-18 model and performs classification
4	Image Processing	OpenCV	Captures video frames and displays detection results
5	GPIO Control	RPi.GPIO	Controls buzzer using Raspberry Pi pins
6	Communication	pySerial	Sends SMS via GSM module using serial communication
7	Data Handling	JSON	Stores detection logs in structured format
8	Development Tools	Thonny / VS Code	Used for writing and debugging code

7. Implementation Visualization



8. Conclusion

- The proposed system provides real-time wild animal detection and alerting using ML models (ResNet18).
- It helps in preventing crop damage, livestock loss, and human-animal conflicts.
- The solution integrates Raspberry Pi, IoT, ML and AI for seamless monitoring.
- Features like customized sound alerts and SMS notifications make it highly practical.
- Applicable not only in agriculture but also in forests, highways, schools, and residential areas near forests.
- The system contributes to smart farming, public safety, and wildlife conservation.

9. References

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THANK YOU